



DOC ID: SL-POP-ERP-MS-003

Consolidated Release: October 2026 • Ver 1.0 Final

MULTI-WAREHOUSE & STORE VERIFICATION ENGINE

Module 3: Store Verification, Stock Control & Location Management

Comprehensive 15-week implementation roadmap for Inward Store Verification (PVN), 60/40 sampling daily stock audits, 5-tier location spatial layout, multi-branch stock transfers (GIT) with handheld QR scanning, and FIFO valuation for **Popular Auto Spare & A/C Parts Co. W.L.L.**

TOTAL MODULE FEE

BD 5,713.636

IMPLEMENTATION TIMELINE

15 Working Weeks

DELIVERY EFFORT

3.75 Engineering Months

MILESTONE TRANCHES

4 Verified Gates (25%)


5 Tiers

Zone/Aisle/Rack/Shelf/Bin



Android PDA

QR Scanner Sync Engine



7-Stage GIT

Branch Transfer Custody



15-Day UAT

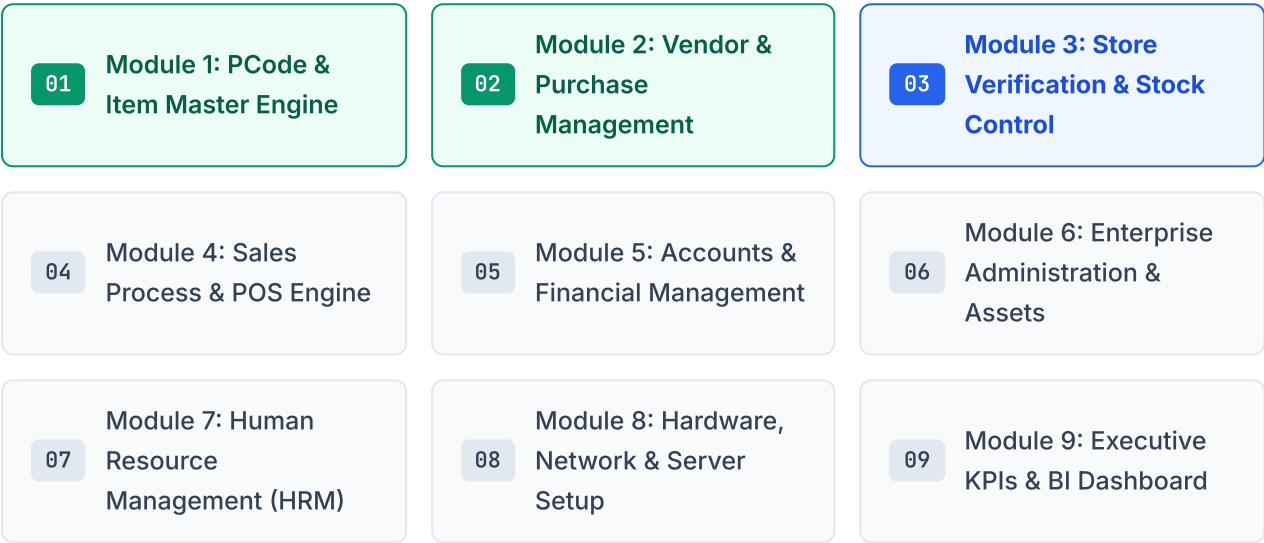
Acceptance SLA

SECTION 1.0

MASTER PLAN
ALIGNMENT

Strategic Position in 9-Module ERP
Architecture

Module 3 operates as the foundational inventory control powerhouse of the Popular ERP ecosystem, interfacing real-time physical bin coordinates, Android PDA barcode readers, daily cycle audits, and inter-branch Goods-in-Transit (GIT) custody with Financials and Sales:



SECTION 2.0

BUSINESS ANALYSIS
SPECIFICATION

Comprehensive Scope Matrix & Technical Architecture (DOC-003 BA Specification)

Synthesized directly from the 40-page Business Analysis specification (DOC-003 v1.0), Module 3 establishes airtight warehouse governance, zero inventory leakage, handheld PDA barcode operations, and real-time multi-branch stock reconciliation:



Inward Store Verification & PVN Generation

Three-way physical verification comparing Supplier Invoices, Goods Received Notes (GRN), and Physical Verification Notes (PVN). Automated barcode scanning logs physical vs invoiced counts, recording instant shortage/damage claims before stock ingestion.



Blind 60/40 Daily Cycle Count & Day Closing

Automated daily audit engine selecting 60% high-velocity/high-value SKUs and 40% randomized items for blind stock counting. Implements day-closing hard locks that prohibit ERP transactions until daily physical count discrepancies are verified.



5-Tier Spatial Location Hierarchy (Zone to Bin)

Standardized spatial coordinate numbering (Warehouse → Zone → Aisle → Rack → Shelf → Bin) with thermal QR label generation. Guides warehouse pickers via the fastest path algorithm to minimize pick times and eliminate wrong part retrievals.



7-Stage Goods-in-Transit (GIT) Branch Transfers

End-to-end custody tracking across 7 status checkpoints (Request → Approval → Pick & Pack → Dispatch → In-Transit → Arrival → Receiving Audit). Handheld driver sign-offs and destination receiving scans ensure zero lost inventory during inter-branch transfers.



Damaged Stock Quarantine & Scrap Disposition

Dedicated Quarantine Virtual Store isolation for returned, damaged, or defective spare parts. Multi-level approval matrix for vendor return credit notes, scrap write-offs, and automated General Ledger write-down postings.

Dynamic Risk Score & Auto-Replenishment Engine

AI-driven inventory risk profiling based on variance frequency, value concentration, and historical shrinkage. Auto-generates min/max replenishment purchase triggers based on 90-day branch velocity and seasonal demand spikes.

SECTION 3.0

15-WEEK RESOURCE
ALLOCATION

Dedicated Engineering Team Cost & Resource
Allocation Matrix (BHD)

Pricing for Module 3 is determined strictly via our transparent, deliverable-based engineering rate matrix across a 15-week development lifecycle (3.75 calendar months) with a dedicated 5-person technical squad:

Project Manager & Enterprise Solutions Architect

BD 545.455 / mo

BD 136.364 / Wk • 3.75 Mo = BD 2,045.456

100% Dedicated

Back-End Lead Engineer (PHP MVC / REST APIs)

BD 227.273 / mo

BD 56.818 / Wk • 3.75 Mo = BD 852.274

100% Dedicated

Front-End Lead Engineer (Admin UI / React)

BD 227.273 / mo

BD 56.818 / Wk • 3.75 Mo = BD 852.274

100% Dedicated

Database & Cloud Infrastructure Architect

BD 204.545 / mo

BD 51.136 / Wk • 3.75 Mo = BD 767.044

100% Dedicated

QA Automation & Multi-Branch Test Lead

BD 159.091 / mo

BD 39.773 / Wk • 3.75 Mo = BD 596.591

100% Dedicated

ENGINEERING DESIGNATION	MONTHLY RATE (BHD)	WEEKLY RATE (BHD)	HOURLY RATE (160H/MO)	EFFORT ALLOCATION	TOTAL COST (BHD)
Project Manager & Enterprise Solutions Architect	BD 545.455	BD 136.364	BD 3.409	3.75 Months (15 Wks)	BD 2,045.456
Back-End Lead Engineer (PHP MVC / REST APIs)	BD 227.273	BD 56.818	BD 1.420	3.75 Months (15 Wks)	BD 852.274

ENGINEERING DESIGNATION	MONTHLY RATE (BHD)	WEEKLY RATE (BHD)	HOURLY RATE (160H/MO)	EFFORT ALLOCATION	TOTAL COST (BHD)
Front-End Lead Engineer (Admin UI / React Tokens)	BD 227.273	BD 56.818	BD 1.420	3.75 Months (15 Wks)	BD 852.274
Database & Cloud Infrastructure Architect	BD 204.545	BD 51.136	BD 1.278	3.75 Months (15 Wks)	BD 767.044
QA Automation & Multi-Branch Test Lead	BD 159.091	BD 39.773	BD 0.994	3.75 Months (15 Wks)	BD 596.591
TOTAL MODULE 3 DEDICATED SQUAD INVESTMENT	BD 1,523.636 / mo	BD 380.909 / wk	BD 8.523 / hr	15 Weeks (3.75 Mo)	BD 5,713.636



Fixed Price Guarantee

Total project investment for Module 3 is fixed at **BD 5,713.636**. No hidden deployment, scanner integration, or hourly surcharges apply within the agreed functional scope.

SECTION 4.0

SPRINT EXECUTION
PLAN

Detailed Milestone Breakdown & Deliverables (15-Week Roadmap)

The implementation is organized into 4 sequential deliverable milestones (25.00% each), guaranteeing focused execution, continuous integration, and transparent acceptance:

1

Milestone 1: 5-Tier Spatial Hierarchy, Bin QR Generator & Inward Store Verification (PVN)

Weeks 1 – 4 (4 Wks)

Setup of centralized warehouse spatial architecture (Warehouse, Zone, Aisle, Rack, Shelf, Bin), dynamic thermal QR code generator for bin labels, and the Inward Store Verification module linked to supplier shipments and Purchase Verification Notes (PVN).

✓ **5-Tier Spatial Layout Matrix**
Warehouse, Zone, Aisle, Rack, Shelf, Bin coordinate hierarchy.

✓ **Bin QR Label Generator** Automated high-density thermal QR label formatting and barcode printing.

✓ **Inward PVN Engine** Physical verification matching against supplier GRN, invoice, and purchase order.

2

Milestone 2: Android PDA Handheld Scanner Engine, Blind 60/40 Cycle Counts & Variance Hard-Lock

Weeks 5 – 8 (4 Wks)

Engineering of Android PDA handheld scanning application with offline data caching, the daily blind cycle audit algorithm (60% high-velocity + 40% random SKUs), and automated day-closing hard locks preventing unverified inventory transactions.

✓ **Android PDA Integration** Scanner event listeners, rapid barcode reading, and offline batch sync.

✓ **Blind 60/40 Cycle Engine** Dynamic audit generation with hidden system stock counts for unbiased scanning.

3

- ✓ **Day Closing Hard-Lock** System-enforced lock preventing sales closing until physical count variances are cleared.

Milestone 3: 7-Stage Goods-in-Transit (GIT) Stock Transfers, Driver Manifest & FIFO Valuation

Weeks 9 – 12 (4 Wks)

Implementation of the 7-stage inter-branch stock transfer lifecycle, digital driver manifest signatures, real-time in-transit inventory ledger tracking, destination receiving variance audits, and FIFO valuation synchronization.

- ✓ **7-Stage GIT Workflow** Request → Approval → Pick & Pack → Dispatch → In-Transit → Arrival → Audit.
- ✓ **Driver Digital Manifest** Electronic dispatch sign-off, vehicle allocation, and route custody tracking.
- ✓ **FIFO Valuation Engine** Real-time landed cost re-calculation and GL transfer ledger postings.

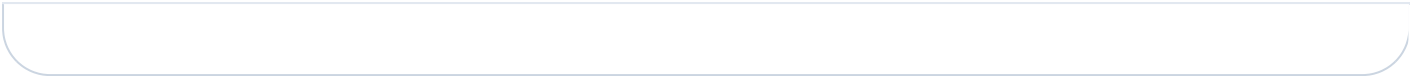
4

Milestone 4: Damaged Stock Quarantine, Auto-Replenishment Min/Max Engine & Final UAT

Weeks 13 – 15 (3 Wks)

Deployment of Quarantine Virtual Store isolation, scrap disposal approvals, supplier return credit note reconciliation, AI risk scoring for shrinkage detection, auto-replenishment min/max triggers, and final multi-branch UAT sign-off.

- ✓ **Damaged Stock Quarantine** Physical isolation, return-to-vendor credit notes, and scrap accounting.
- ✓ **Min/Max Replenishment** Velocity-based automated reorder generation and safety stock alerts.
- ✓ **Multi-Warehouse UAT** Live scanner testing across central warehouse and branches, followed by final sign-off.



SECTION 5.0

DELIVERABLE-BASED
INVOICING

Milestone Payment Structure & Invoicing Schedule (15-Week Model)

Payments are structured in 4 equal milestone disbursements of **25.00% (BD 1,428.409 each)**, tied strictly to formal milestone deliverables and client sign-off:

MILESTONE PHASE	DELIVERABLE DESCRIPTION	TIMELINE	PERCENTAGE	AMOUNT (BHD)	PAYMENT TRIGGER & ACCEPTANCE CRITERIA
Milestone 1	5-Tier Spatial Hierarchy, Bin QR Generator & Inward Store Verification (PVN)	Weeks 1–4	25.00%	BD 1,428.409	Demonstration of Spatial Master, Bin QR generator & Inward PVN matching
Milestone 2	Android PDA Handheld Scanner Engine, Blind 60/40 Cycle Counts & Variance Locks	Weeks 5–8	25.00%	BD 1,428.409	Sign-off of Android PDA scanner integration, 60/40 blind audits & day-closing locks
Milestone 3	7-Stage Goods-in-Transit (GIT) Stock Transfers, Driver Manifest & FIFO Valuation	Weeks 9–12	25.00%	BD 1,428.409	Verification of 7-stage GIT transfers, driver sign-off & FIFO landed cost math
Milestone 4	Damaged Stock Quarantine, Auto-Replenishment Min/Max Engine & Final UAT	Weeks 13–15	25.00%	BD 1,428.409	Delivery of Quarantine store, Auto-replenishment engine & Final multi-branch UAT sign-off
Total Milestone	Complete Module 3 ERP Scope Delivery	15 Weeks	100.00%	BD 5,713.636	Formal Acceptance & Sign-off

MILESTONE PHASE	DELIVERABLE DESCRIPTION	TIMELINE	PERCENTAGE	AMOUNT (BHD)	PAYMENT TRIGGER & ACCEPTANCE CRITERIA
Project Cost					

SECTION 6.0

GOVERNANCE
FRAMEWORK

Working Calendar, SLA, 15-Day Grace Period & Legal Terms



Working Calendar & Team Hours

Development operates on a 5-day standard work week (Sunday to Thursday, 09:00 to 18:00 AST), aligning with GCC financial business hours. Sprint demos are scheduled at the end of each milestone phase.



15-Day Milestone Review & Grace Period

Client is granted a formal 15-calendar-day review window following each sprint deployment to verify warehouse scanning, inventory transfers, and audit tracking before milestone invoice closure.



SLA & Defect Severity Matrix

Critical Stock Discrepancy & Day-Closing Blockers (Severity 1) resolved within 4 hours; Major Transfer Calculation Defects (Severity 2) within 24 hours; Minor UI adjustments (Severity 3) within 3 business days.



Scope Control & Document Security

Features adhere strictly to the 40-page specifications in DOC-003. All stock transfer manifests and inventory audit records are encrypted with AES-256 for corporate confidentiality.

FINAL STAGE

OFFICIAL ACCEPTANCE & DIGITAL VERIFICATION

Stakeholder Authorization & Digital Sign-Off Console

By affixing digital signatures and authorizations below, all participating executive and technical stakeholders ratify the deliverables, implementation timeline, SLA commitments, and financial disbursement gates for **Module 3: Store Verification & Stock Control (SL-POP-ERP-MS-003)**:

1. Engineering Preparation & Multi-Tier Verification Matrix

Verification Stage	Responsible Party	Signatory & Designation	Date	Status
Prepared By	SaNDS Lab Middle East W.L.L	Ancy Varghese Thekkan (Software Architect)	27/02/2026	Completed & Verified
Verified By	SaNDS Lab Middle East W.L.L	Ajit Kumar KV (CEO & Lead Solutions Architect)	05/03/2026	Checked & Ratified
Consultant Review	UniGlobal Consultancy	Lead Enterprise Strategic Advisor	05/03/2026	PENDING VERIFICATION
Client Approval	Popular Auto Spare Co. W.L.L	Managing Director / Executive Commercial Sponsor	October 2026	PENDING SIGN-OFF

2. Executive Multi-Party Digital Acceptance Console

Popular Auto Spare Co. W.L.L

Client Executive Sign-off & Commercial Approval

No Signature on Record

PENDING SIGN-OFF

SaNDS Lab Middle East W.L.L

Lead Solutions Architect & Engineering Director

 No Signature on Record

PENDING SIGN-OFF

UniGlobal Consultancy

Executive Strategic Advisor & Governance Lead

 No Signature on Record

PENDING SIGN-OFF